

Spectral Metrics of Transformer Training Are Determined by a Single Direction

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Abstract

Studies of transformer pretraining read training dynamics off spectral summaries of hidden states: the leading eigenvalue share E_1 , effective rank, RankMe, CKA, and anisotropy. We show that one direction fixes the value of each, through an identity $M(\mathbf{H}) = e^{h_b(t)} M(\mathbf{C})^{1-t}$, where $\mathbf{C}$ is the residual after that direction is projected out and t is the direction’s share of the spectrum M is built from. The identity holds to 3×10^{-15} relative error on 4,743 measurements covering Pythia at 160M, 410M and 1.4B, OLMo-2 1B, every layer, 20 checkpoints and three disjoint corpus samples. Effective rank is built from eigenvalues and RankMe from singular values, so the same direction enters the two with different shares. We prove the RankMe share never exceeds the effective-rank share, and it does not in any of the 4,743 records. The split this predicts is the split the data shows. In Pythia-1.4B the direction holds 85 percent of the eigenvalue mass and 6 percent of the singular-value mass. Effective rank falls $105\times$ over training and grows $2.1\times$ once the direction is removed. RankMe never collapses, and still reverses sign or understates the residual trend by $9\times$. The direction is the attention sink. It carries half of its energy on the 0.2 percent of tokens at sequence position 0, and its strength alone predicts E_1 with $R^2 = 0.94$ to 0.97 . OLMo-2 1B forms no sink in its first 76B tokens, and there raw and residual metrics agree to 3.5 percent. Effective rank has not converged in sample size at 524,288 tokens, so we report ratios at fixed n and never an absolute rank.

1 Introduction

A growing literature reads the training dynamics of transformers off spectral summaries of their hidden states. Li et al. (2025) report three geometric phases in pretraining, measured with RankMe and a power-law spectral exponent. Ethayarajh (2019) measure anisotropy, Timkey & van Schijndel (2021) isolate rogue dimensions, Garrido et al. (2023) propose RankMe as a proxy for downstream quality, and Raghu et al. (2021) compare layers with CKA. Each quantity is a functional of the spectrum of one $n \times d$ matrix of token representations, and each has been read as a description of how a model organizes information.

A separate literature reports that transformer hidden states contain a small number of coordinates with variance orders of magnitude above the rest. Xiao et al. (2024) named the attention sink, Sun et al. (2024) measured massive activations, Gu et al. (2025) traced when sinks appear during training, and Queipo-de Llano et al. (2025) proved that massive activations force the effective rank of a layer down. The two literatures use the same matrices and rarely cite each other.

This paper asks what the first literature measures when the second one applies.

Problem. Fix a checkpoint, a layer, and a corpus of n tokens, and let $\mathbf{H} \in \mathbb{R}^{n \times d}$ hold the hidden states. Let $M(\mathbf{H})$ be a spectral summary. Given a trajectory $\{M(\mathbf{H}_t)\}$ over training steps t , decide which features of the trajectory describe the distribution of the n representations and which describe the size of a few outlier coordinates.

The answer is an identity. Write $\mathbf{s}$ for the top eigenvector of the covariance of $\mathbf{H}$ and $\mathbf{C} = \mathbf{H}(\mathbf{I} - \mathbf{s}\mathbf{s}^\top)$ for the residual. For any statistic M of the form “exponential entropy of a normalized spectrum”,

$$M(\mathbf{H}) = e^{h_b(t)} \cdot M(\mathbf{C})^{1-t}, \quad (1)$$

where h_b is the binary entropy in nats and t is the share of the spectrum carried by $\mathbf{s}$. Equation 1 holds for every matrix. A study that reports effective rank falling from 442 to 4.2 has reported a value of t and a value of $M(\mathbf{C})$ in disguise.

Which t enters depends on the statistic. Effective rank normalizes eigenvalues, so $t = E_1 = \lambda_1 / \sum_i \lambda_i$. RankMe normalizes singular values, so $t = \sigma_1 / \sum_i \sigma_i$. We prove that the second is never larger than the first (Lemma 2), which turns Equation 1 into a prediction: the same direction should wreck effective rank and leave RankMe nearly intact. Section 5 shows that it does, and that the residue in RankMe is still enough to flip the sign of its training trajectory.

Contributions.

1. Effective rank, RankMe, anisotropy, and CKA are determined by one direction through Equation 1 and its analogues (Section 3). The effective-rank instance reproduces measurement to 3×10^{-15} relative error on 4,743 records across four model families. The raw-to-residual ratio is bounded above by the residual rank, which grows with model width, so distortion factors reported in different models are not comparable.
2. The RankMe share never exceeds the effective-rank share (Lemma 2, no violations in 4,743 records). At the peak layer of Pythia-1.4B the two shares are 0.853 and 0.058, and the resulting distortions are $243\times$ and $1.24\times$ (Section 3).
3. In Pythia at 160M, 410M and 1.4B the direction carries 44 to 57 percent of its energy on the 1/512 of tokens at position 0, up from 0.1 percent at initialization, and its strength alone predicts E_1 with $R^2 = 0.94$ to 0.97 (Section 4).
4. Raw effective rank falls by $21\times$ to $119\times$ over training at the three scales; the residual grows by $1.4\times$ to $2.1\times$. Raw RankMe falls by 9 to 23 percent after the sink forms while the residual rises. One early transient, present before the sink exists and reproduced in OLMo-2, survives the projection (Section 5).
5. OLMo-2 1B forms no sink in its first 76B tokens, and there raw and residual metrics agree to a median of 3.5 percent. The gap between raw and residual tracks sink strength rather than the projection itself (Section 5).

The propositions in Section 3 are elementary. Their use is to remove a class of published measurements from the space of empirical findings, to say which residual quantity a study should report instead, and to say which of the five statistics is worth reporting at all.

2 Setting

Hidden states. Fix a model, a layer, and a corpus. Running the corpus through the model yields n token representations in $\mathbb{R}^d$, stacked as the rows of $\mathbf{H} \in \mathbb{R}^{n \times d}$. Write $\bar{\mathbf{H}} = (\mathbf{I}_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top)\mathbf{H}$ for the column-centered matrix, $\Sigma = \frac{1}{n}\bar{\mathbf{H}}^\top\bar{\mathbf{H}}$ for the covariance, and $\lambda_1 \geq \dots \geq \lambda_d \geq 0$ for its eigenvalues with eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_d$. Set $V = \text{tr } \Sigma$ and $\sigma_i = \sqrt{n\lambda_i}$ for the singular values of $\bar{\mathbf{H}}$.

Spectral summaries. For a nonzero $\mathbf{w} \in \mathbb{R}_{\geq 0}^m$ define the exponential entropy

$$\text{H}_{\text{exp}}(\mathbf{w}) = \exp\left(-\sum_{i=1}^m \tilde{w}_i \log \tilde{w}_i\right), \quad \tilde{w}_i = w_i / \sum_j w_j, \quad (2)$$

with $0 \log 0 = 0$. The five summaries this paper studies are

$$E_1(\mathbf{H}) = \lambda_1/V, \quad \text{ER}(\mathbf{H}) = \text{H}_{\text{exp}}(\lambda_1, \dots, \lambda_d), \quad \text{RankMe}(\mathbf{H}) = \text{H}_{\text{exp}}(\sigma_1, \dots, \sigma_d), \quad (3)$$

$$\rho(\mathbf{H}) = \frac{1}{n(n-1)} \sum_{i \neq j} \frac{\langle \mathbf{H}_i, \mathbf{H}_j \rangle}{\|\mathbf{H}_i\| \|\mathbf{H}_j\|}, \quad \text{CKA}(\mathbf{X}, \mathbf{Y}) = \frac{\|\mathbf{X}^\top \mathbf{Y}\|_F^2}{\|\mathbf{X}^\top \mathbf{X}\|_F \|\mathbf{Y}^\top \mathbf{Y}\|_F}. \quad (4)$$

Here $\mathbf{H}_i$ is the i -th row of $\mathbf{H}$, ρ is the anisotropy of Ethayarajh (2019), and CKA is linear centered kernel alignment (Kornblith et al., 2019) for column-centered $\mathbf{X} \in \mathbb{R}^{n \times d_1}$ and $\mathbf{Y} \in \mathbb{R}^{n \times d_2}$. Effective rank (Roy & Vetterli, 2007) and RankMe (Garrido et al., 2023) apply the same functional to different inputs: the training-dynamics literature feeds H_{exp} normalized eigenvalues and the self-supervised-learning literature feeds it normalized singular values. We keep both conventions and label them separately, because Section 3 shows the choice decides how much damage one direction can do.

The projection. Define

$$\mathbf{C} = \mathbf{H}(\mathbf{I}_d - \mathbf{s}\mathbf{s}^\top), \quad \mathbf{s} = \mathbf{v}_1. \quad (5)$$

We call $\mathbf{C}$ the *residual* and $\mathbf{s}$ the *leading direction*. Removing $\mathbf{s}$ from a matrix of embeddings is the All-but-the-Top postprocessing of Mu & Viswanath (2018) with one component; Li et al. (2020) and Rajaei & Pilehvar (2021) use related corrections to improve downstream sentence similarity. This paper uses the projection as a measurement control rather than as a way to improve a downstream score.

Sink strength. For a corpus of B sequences of length T , let α be the mean norm of the hidden states at sequence position 0 divided by the mean norm at all other positions. An attention sink at this layer makes α large (Xiao et al., 2024; Sun et al., 2024). We also report the *position-0 mass* of the leading direction, the fraction of $\sum_i \langle \mathbf{H}_i, \mathbf{s} \rangle^2$ contributed by rows at position 0.

Models, corpora, and protocol. The training-dynamics measurements use Pythia at 160M, 410M and 1.4B with the 20 released checkpoints from step 0 to step 143,000 (Biderman et al., 2023), and the 13 released stage-1 checkpoints of OLMo-2 1B (OLMo Team et al., 2025). Every checkpoint sees the same 606,208 OpenWebText tokens (Gokaslan & Cohen, 2019) in sequences of 512. We split them into three disjoint blocks of 200,704 tokens and treat the blocks as independent samples. The token array carries 28,552 distinct ids, no duplicated sequences, and a published SHA-256 prefix. Weights are loaded in float32 and all layers are measured. Cross-block spread at the peak layers is 0.1 to 1.9 percent, and every figure shows it. Section 6 adds final-checkpoint measurements of GPT-2 and GPT-2-XL (Radford et al., 2019), Pythia-6.9B, and Qwen2.5-1.5B (Yang et al., 2024) on WikiText-2 (Merity et al., 2017) and OpenWebText from an earlier round; Table 8 lists the corpus and token count behind every number in the paper. Total compute is under 40 GPU-hours.

3 One direction fixes the value of each metric, and by how much

3.1 An entropy split and an ordering

Lemma 1 (Top-entry split). *Let $\mathbf{w} \in \mathbb{R}_{\geq 0}^m$ with $\|\mathbf{w}\|_1 > 0$, let $w_1 = \max_i w_i$, let $t = w_1/\|\mathbf{w}\|_1 < 1$, and let $\mathbf{w}_{-1}$ be $\mathbf{w}$ with its first entry deleted. Then*

$$H_{\text{exp}}(\mathbf{w}) = e^{h_b(t)} \cdot H_{\text{exp}}(\mathbf{w}_{-1})^{1-t}, \quad h_b(t) = -t \log t - (1-t) \log(1-t). \quad (6)$$

Lemma 1 restates the chain rule for Shannon entropy. Everything below follows from it and from one observation: centering acts on the left and the projection in Equation 5 acts on the right, so $\bar{\mathbf{C}} = \bar{\mathbf{H}}(\mathbf{I} - \mathbf{s}\mathbf{s}^\top)$, the covariance of $\mathbf{C}$ equals $\Sigma - \lambda_1 \mathbf{s}\mathbf{s}^\top$, and its spectrum is $\{\lambda_2, \dots, \lambda_d, 0\}$.

Lemma 2 (Share ordering). *Let $p = \lambda_1/\sum_i \lambda_i$ and $r = \sigma_1/\sum_i \sigma_i$ with $\sigma_i = \sqrt{n\lambda_i}$. Then*

$$\frac{p}{r} = \frac{\sigma_1 \sum_i \sigma_i}{\sum_i \sigma_i^2} \geq 1, \quad (7)$$

with equality only when the spectrum has a single nonzero entry.

3.2 The identities

Proposition 1 (Leading share). *Let $x = \lambda_1 / \text{tr}(\mathbf{\Sigma} - \lambda_1 \mathbf{s} \mathbf{s}^\top)$. Then $E_1(\mathbf{H}) = x/(x+1)$, a strictly increasing bijection from $[0, \infty)$ onto $[0, 1)$. The value of $E_1(\mathbf{H})$ carries exactly the ratio of leading to residual variance, and nothing about the shape of the residual spectrum.*

Proposition 2 (Effective rank). *With $p = E_1(\mathbf{H}) < 1$,*

$$\text{ER}(\mathbf{H}) = e^{h_b(p)} \cdot \text{ER}(\mathbf{C})^{1-p}. \quad (8)$$

Proposition 3 (RankMe). *With $r = \sigma_1 / \sum_i \sigma_i$, $\text{RankMe}(\mathbf{H}) = e^{h_b(r)} \cdot \text{RankMe}(\mathbf{C})^{1-r}$.*

Corollary 1 (Distortion, and which metric it reaches). *Write $D_M = M(\mathbf{C})/M(\mathbf{H})$. Then $D_M = M(\mathbf{C})^t e^{-h_b(t)}$ with t the share belonging to M , so*

$$D_M \leq M(\mathbf{C}), \quad D_M \rightarrow M(\mathbf{C}) \text{ as } t \rightarrow 1, \quad (9)$$

and $D_M > 1$ exactly when $t \log M(\mathbf{C}) > h_b(t)$. Since $r \leq p$ by Lemma 2 and D increases in t once $D > 1$, RankMe is never distorted more than effective rank at comparable residual ranks.

Corollary 1 settles how to read a distortion figure. A study that removes the leading direction from a $d = 4096$ model and reports a $700\times$ change in effective rank has reported a large p and a residual rank near 10^3 . The same p in a $d = 512$ model cannot produce a factor above a few hundred, because D never exceeds $\text{ER}(\mathbf{C})$. Cross-model comparisons of distortion factors compare model width as much as they compare contamination.

Proposition 4 (Anisotropy). *Let $\hat{\mathbf{u}}_i = \mathbf{H}_i / \|\mathbf{H}_i\|$ and $\beta_i = \langle \hat{\mathbf{u}}_i, \mathbf{s} \rangle$. Then*

$$\rho(\mathbf{H}) = \frac{\|\sum_i \hat{\mathbf{u}}_i\|^2 - n}{n(n-1)} = \frac{(\sum_i \beta_i)^2 + \|\sum_i \hat{\mathbf{u}}_i^\perp\|^2 - n}{n(n-1)}, \quad (10)$$

where $\hat{\mathbf{u}}_i^\perp = \hat{\mathbf{u}}_i - \beta_i \mathbf{s}$. Removing $\mathbf{s}$ deletes the first term. That term reaches n^2 when the β_i share a sign and magnitude, and it is one of two, so the decomposition bounds the sink’s contribution without predicting that it dominates.

Proposition 5 (CKA homogenization). *Let $\mathbf{X} = \mathbf{u} \mathbf{p}^\top + \mathbf{X}_c$ and $\mathbf{Y} = \mathbf{u} \mathbf{q}^\top + \mathbf{Y}_c$ be column-centered with $\mathbf{X}_c^\top \mathbf{u} = \mathbf{Y}_c^\top \mathbf{u} = \mathbf{0}$, and set $A = \|\mathbf{u}\|^2 \|\mathbf{p}\|^2$, $B = \|\mathbf{u}\|^2 \|\mathbf{q}\|^2$. Then*

$$1 - \text{CKA}(\mathbf{X}, \mathbf{Y}) = O(A^{-1/2} + B^{-1/2}), \quad (11)$$

with constants depending only on $\|\mathbf{X}_c^\top \mathbf{Y}_c\|_F$, $\|\mathbf{X}_c^\top \mathbf{X}_c\|_F$, and $\|\mathbf{Y}_c^\top \mathbf{Y}_c\|_F$. Two layers sharing one high-norm rank-one term have $\text{CKA} \rightarrow 1$ whatever their residuals do.

Proposition 6 (Specificity of the leading direction). *Removing the k -th eigendirection, $\mathbf{H}_{-k} = \mathbf{H}(\mathbf{I} - \mathbf{v}_k \mathbf{v}_k^\top)$, gives $E_1(\mathbf{H}_{-k}) = \lambda_1 / (V - \lambda_k) \geq E_1(\mathbf{H})$ for $k \geq 2$. For $\mathbf{u}$ drawn uniformly on the unit sphere, $\mathbb{E}[\mathbf{u}^\top \mathbf{\Sigma} \mathbf{u}] = V/d$, so a random direction removes $1/d$ of the total variance in expectation.*

3.3 The identity is exact and the ordering is never violated

Figure 1(a) evaluates Equation 8 on every record. The median relative error is 5.3×10^{-16} and the largest is 2.9×10^{-15} , which is the accumulated rounding of a double-precision eigendecomposition. Panel (b) plots the two shares against each other. Lemma 2 is never violated, the median p/r is 6.2, and the converged Pythia checkpoints sit at the bottom right, where p approaches 1 while r stays near 0.06.

Table 1 reads the split off the converged checkpoints. At Pythia-1.4B the leading direction holds 85.3 percent of the eigenvalue mass and 5.8 percent of the singular-value mass. Feeding those two shares through Equations 8 and the RankMe analogue gives $243\times$ and $1.24\times$, and both match measurement. Squaring the spectrum is what separates the two statistics: it moves mass onto the leading entry, and a statistic that squares first is the one a single direction can capture.

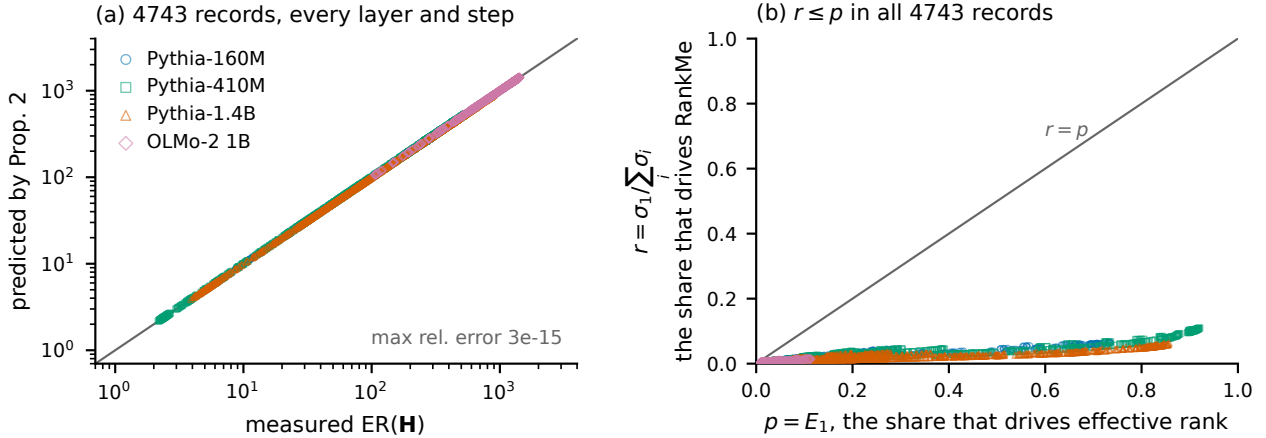


Figure 1: **The value of a rank statistic is fixed by one share and the residual.** (a) Effective rank predicted by Equation 8 against the measured value, for all 4,743 (model, layer, checkpoint, block) records: Pythia at 160M, 410M and 1.4B over 20 checkpoints and every layer, OLMo-2 1B over 13, three disjoint corpus blocks each. The largest relative error is 2.9×10^{-15} , at the level of double precision. (b) The two shares. Effective rank is driven by $p = E_1$ and RankMe by $r = \sigma_1 / \sum_i \sigma_i$. Lemma 2 forces $r \leq p$; the median ratio is 6.2 and the largest is 20.6. Points near $p = 1$ and $r \approx 0.06$ are the converged Pythia checkpoints, where the same direction destroys effective rank and barely moves RankMe.

Table 1: **The same direction, two statistics, two outcomes.** Peak-sink layer, step 143,000, mean over three disjoint corpus blocks. p and r are the shares of Propositions 2 and 3; r is recovered by inverting the identity. D is the raw-to-residual ratio. The final column is the bound of Corollary 1 for effective rank.

Model	Layer	d	$p = E_1$	r	p/r	D_{ER}	D_{RankMe}	$ER(\mathbf{C})$
Pythia-160M	L4 of 12	768	0.696	0.061	11.5	$32\times$	$1.18\times$	356
Pythia-410M	L9 of 24	1024	0.913	0.104	8.8	$235\times$	$1.45\times$	548
Pythia-1.4B	L5 of 24	2048	0.853	0.058	14.7	$243\times$	$1.24\times$	1023

Takeaway. A reported change in effective rank, RankMe, or the raw-to-residual ratio adds nothing to a report of the corresponding share and the residual value. Sections 4 and 5 therefore ask what makes the share move.

4 The leading direction is the attention sink

Section 3 holds for any matrix. It becomes a statement about transformers once we identify $\mathbf{s}$.

The direction ends up on one token position. Table 2 tracks three quantities across checkpoints. At initialization the leading direction places 0.1 percent of its mass on position 0 and α sits near 1.2; the direction is ordinary content variance, and $E_1(\mathbf{H})$ equals $E_1(\mathbf{C})$ to two digits. By step 143,000 it carries 44 to 57 percent of its mass on the 0.2 percent of tokens at position 0, a 250-fold enrichment, and α reaches 19.5 to 44.7. Over the same interval $E_1(\mathbf{C})$ stays between 0.028 and 0.048. The residual spectrum stays diffuse at every checkpoint, which is what makes the projection in Equation 5 a clean split rather than a rebalancing of two comparable components.

A spectral gap marks the layers where the identification holds. The ratio σ_1/σ_2 is 14.7, 14.9, and 14.6 at GPT-2 layers L2, L3, and L4, and falls to 1.1 at L0 and 3.0 at L11. In Qwen2.5-1.5B it reaches 154 at L2. Where the gap is near 1 the leading direction is not separated from content directions and $\mathbf{s}$ should not be read as the sink. We report the gap alongside every layer-level claim and treat $\sigma_1/\sigma_2 > 3$ as the condition under which the identification is meaningful.

Table 2: **The leading direction migrates onto position 0 during training.** Peak-sink layer, mean over three disjoint blocks of 200,704 OpenWebText tokens. Position 0 is $1/512 = 0.2$ percent of the corpus. The residual share $E_1(\mathbf{C})$ is flat from random initialization to convergence in all three models, so the entire E_1 trajectory lives in one direction.

Model	Layer	α		position-0 mass		$E_1(\mathbf{C})$	
		step 0	step 143k	step 0	step 143k	step 0	step 143k
Pythia-160M	L4	1.23	19.5	0.001	0.441	0.041	0.048
Pythia-410M	L9	1.20	44.7	0.001	0.569	0.036	0.028
Pythia-1.4B	L5	1.22	25.8	0.001	0.498	0.038	0.033
OLMo-2 1B	L3	1.01	1.18 [†]	0.008	0.002 [†]	0.015	0.008 [†]

[†]OLMo-2 stage-1 step 36,000, which is 76B tokens and about 2 percent of its run.

Table 3: **Removing the leading direction is not removing a direction.** Effective rank at the peak-sink layer, step 143,000, same corpus. The random column is the mean and standard deviation over five uniformly drawn unit directions. Proposition 6 predicts the PC2, PC3, and random columns.

Model	Layer	baseline	remove PC1	remove PC2	remove PC3	remove random
Pythia-160M	L4	11.0	352.1	10.5	10.6	10.9 ± 0.01
Pythia-410M	L9	2.3	539.8	2.3	2.3	2.3 ± 0.00

No other direction does anything. Table 3 removes PC1, PC2, PC3, and five random unit directions from the same matrices. At Pythia-410M L9, removing PC1 takes effective rank from 2.3 to 539.8; removing PC2, PC3, or a random direction leaves it at 2.3. This run uses an independent implementation from the one behind Table 2, and at Pythia-160M L6 the two agree to every printed digit on the block they share (11.0 and 223.2).

Takeaway. At layers with a large spectral gap the leading direction is the attention sink, it becomes so during training rather than at initialization, and no other single direction reproduces its effect.

5 Sink growth accounts for the reported trajectories

5.1 Two phenomena, one of which survives the projection

Figure 2 tracks the peak-sink layer of each Pythia model. Two features of the raw effective-rank curve have been read as training phases: a drop and recovery before step 1,000, and a long decline after step 4,000.

The early transient survives the projection. Residual effective rank falls from 478 to 195 between step 16 and step 256 at Pythia-1.4B, from 295 to 90 at Pythia-410M, and from 250 to 114 at Pythia-160M, then recovers. Sink strength stays between 1.05 and 1.23 throughout that window, so the sink does not exist yet. Residual E_1 rises over the same steps, to 0.096 at 1.4B and 0.160 at 410M, and returns below 0.03 by step 2,000. Representations reorganize here, and the projection does not undo it.

The later decline does not survive. Raw effective rank falls from 442 to 4.2 at Pythia-1.4B, 279 to 2.3 at 410M, and 237 to 11.1 at 160M, a factor of 105, 119, and 21. The residual grows over the same interval, from 478 to 1023, 295 to 548, and 250 to 356, a factor of 2.1, 1.9, and 1.4. All three residual curves rise, and all three raw curves fall.

5.2 RankMe does not collapse and still points the wrong way

The bottom row of Figure 2 runs RankMe on the same matrices. Nothing collapses: raw RankMe moves by -14 to $+3$ percent across all of training while raw effective rank moves by two orders of magnitude.

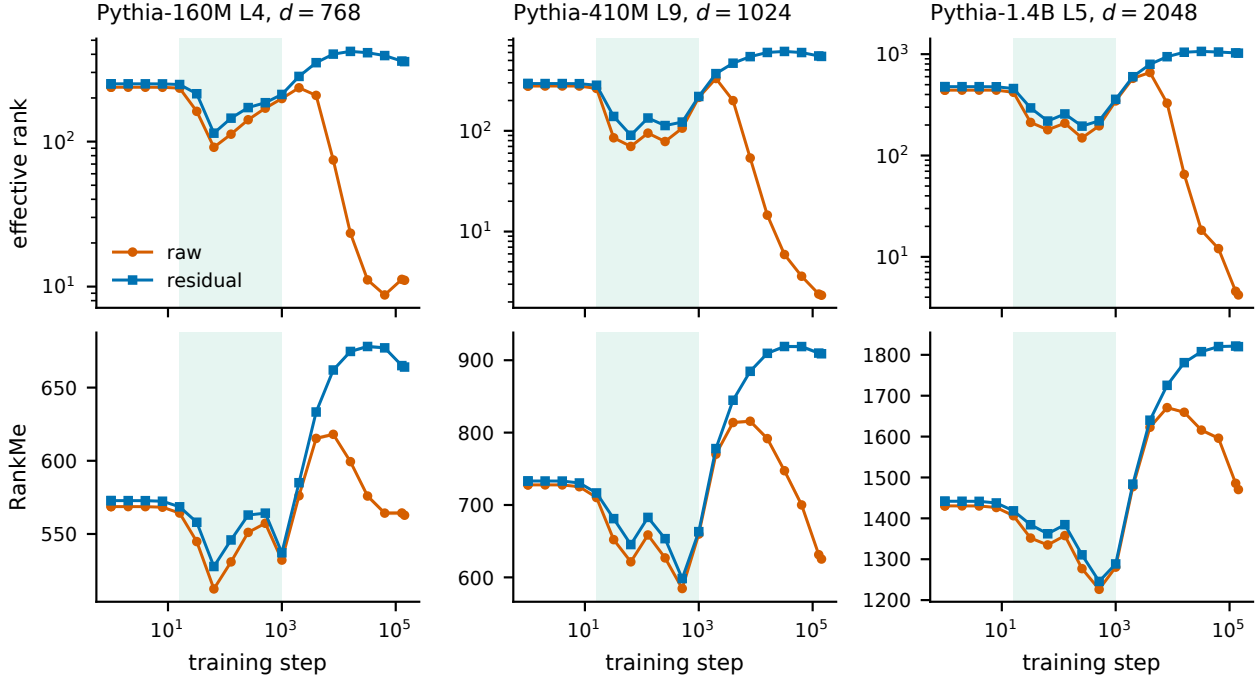


Figure 2: **The collapse is the sink; the early transient is not.** Peak-sink layer of each model, 20 released checkpoints, mean over three disjoint blocks of 200,704 OpenWebText tokens with the block spread shaded. Shading in green marks steps 16 to 1,000. **Top:** effective rank. Raw falls by 21 $\times$ to 119 $\times$; the residual dips early, recovers, and ends above where it started. **Bottom:** RankMe on the same matrices. Neither curve collapses, and the raw curve turns down after the sink forms while the residual keeps rising.

Corollary 1 predicts this, because the RankMe share is 0.06 to 0.10 where the effective-rank share is 0.70 to 0.91.

The residue is still large enough to invert the reading. Over all of training the raw curve moves -1.0 percent at 160M, -14.1 percent at 410M, and $+2.8$ percent at 1.4B, while the residual moves $+15.9$, $+24.0$, and $+26.2$ percent. Two of the three raw trajectories have the wrong sign, and the third understates the residual trend by a factor of 9.4. Restricting to the window where the sink actually grows, step 8,000 to step 143,000, the raw curve falls by 8.9, 23.3, and 12.0 percent while the residual changes by $+0.3$, $+2.7$, and $+5.5$ percent. A RankMe trajectory read at face value reports a late decline that the residual does not have.

5.3 Sink strength predicts the leading share

Proposition 1 reduces E_1 to the ratio x of leading to residual variance. A rank-one term of magnitude α contributes variance proportional to α^2 , which predicts $x = \kappa\alpha^2$ and

$$E_1(\mathbf{H}) = \frac{\kappa\alpha^2}{\kappa\alpha^2 + 1}. \quad (12)$$

Fitting the single parameter κ over the 20 checkpoints of each model, with α and E_1 measured on the same tokens, gives $R^2 = 0.951$, 0.939, and 0.971 at 160M, 410M, and 1.4B. The implied x correlates with α^2 at Pearson $r = 0.978$, 0.996, and 0.988. One scalar, measured from token norms and blind to the spectrum, accounts for most of a trajectory that has been read as representational reorganization.

5.4 Injecting a sink into residual states reproduces the trajectory

The fits above show that α tracks E_1 . To test the direction of the relation we start from Pythia-70M at step 512, where $\alpha = 1.24$ and the sink has not formed. We remove the leading direction to obtain $\mathbf{C}$ with

Table 4: **A synthetic rank-one term walks both metrics across their range.** Pythia-70M step 512 layer 3, starting from residual states with $E_1 = 0.063$ and $\text{ER}(\mathbf{C}) = 139.9$. The last row applies Equation 8 with $\text{ER}(\mathbf{C})$ held at 139.9. Agreement begins at $\alpha = 3$, where the injected term overtakes the largest content direction; below that the leading direction is still content and 139.9 is the wrong reference.

Injected α	0	1	2	3	5	7	10	15	20	30
E_1	0.063	0.062	0.060	0.081	0.197	0.325	0.496	0.688	0.797	0.898
ER measured	139.9	140.8	136.3	124.1	86.8	52.8	24.2	8.7	4.5	2.3
ER from Eq. 8	129.7	129.9	130.4	124.1	86.8	52.8	24.2	8.7	4.5	2.3

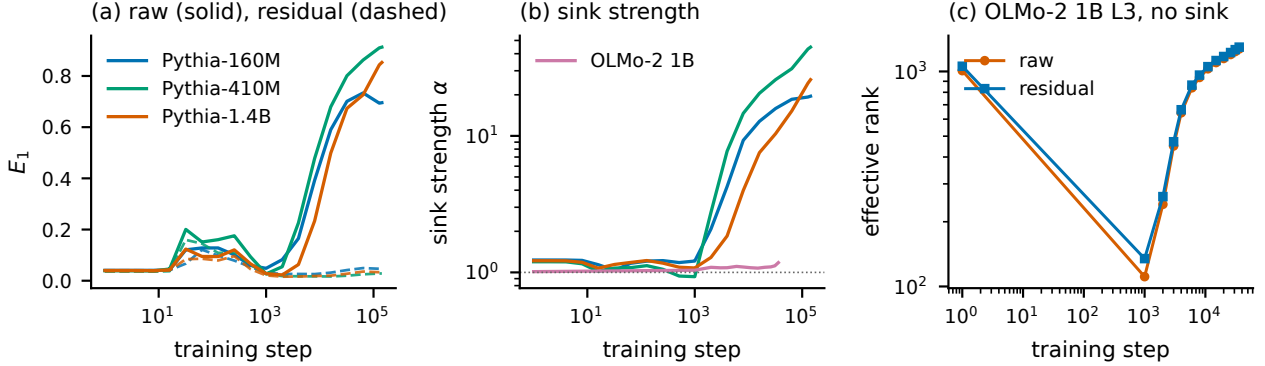


Figure 3: **The raw-to-residual gap is a function of sink strength.** (a) E_1 raw (solid) and residual (dashed) at the peak-sink layer of each Pythia model. The raw curves separate from the residual exactly when α leaves 1. (b) Sink strength. OLMo-2 1B stays below 1.27 across all 17 layers and 13 checkpoints. (c) OLMo-2 1B L3. With no sink, raw and residual effective rank agree to a median of 3.5 percent and both rise. The dip at step 1,000 is the sink-free early transient of Section 5, here at $\alpha = 1.04$.

$E_1 = 0.063$ and $\text{ER} = 140$, then add back a rank-one term $\alpha \mathbf{e}_0 \mathbf{s}^\top$ of controlled magnitude. Table 4 reports the sweep: as α grows from 3 to 30, E_1 rises from 0.08 to 0.90 and effective rank falls from 124 to 2.3, covering the range the raw trajectories traverse over training. The content matrix is fixed, so the entire excursion comes from the injected term. For $\alpha \leq 2$ the injected term is smaller than the largest content direction, so the leading eigenvector is still a content direction and 139.9 is the wrong reference. The scope rule of Section 4 shows up here as an eight percent prediction error.

5.5 A model without a sink shows no gap

Every claim above compares a raw quantity with one computed after a projection. A reader may ask how much of the gap the projection itself creates. OLMo-2 1B answers this without an intervention: over its first 76B tokens it never develops a sink. Sink strength stays below 1.273 across all 17 layers and 13 checkpoints, and the position-0 mass of the leading direction stays below 0.012. On those matrices raw and residual effective rank differ by a median of 3.5 percent, and raw effective rank rises monotonically from 111 to 1272 after the early dip. The projection removes a direction in OLMo-2 exactly as it does in Pythia; what changes is that the direction is not carrying anything.

OLMo-2 also reproduces the early transient. Its effective rank falls from 1008 at initialization to 111 at step 1,000 with $\alpha = 1.04$, then climbs for the rest of the measured run. The sink-free rank dip appears in two architecture families.

We do not read this as evidence that OLMo-2 never forms a sink. Step 36,000 is about 2 percent of its schedule, Pythia’s sink at these layers only becomes visible after a comparable fraction, and OLMo-2’s α is drifting up at the last measured checkpoints. The claim supported here is that OLMo-2 has no sink over the interval we measured, and that where there is no sink there is no gap.

Table 5: **Five summaries at the layer where E_1 peaks.** 200,000 OpenWebText tokens, final checkpoints. Raw values on the left of each arrow, residual on the right. “Rogue” is the share of variance in the top five coordinates (Timkey & van Schijndel, 2021); “sink mass” is the fraction of the leading direction’s ℓ_1 mass carried by those five coordinates.

Model	Layer	E_1	Rogue	sink mass	RankMe	Anisotropy
GPT-2 (124M)	L3	93.6% $\rightarrow$ 11.5%	94.4% $\rightarrow$ 18.4%	88.2%	352 $\rightarrow$ 612	0.550 $\rightarrow$ 0.481
	L12	82.2% $\rightarrow$ 48.5%	94.0% $\rightarrow$ 70.0%	56.0%	191 $\rightarrow$ 346	0.829 $\rightarrow$ 0.556
Pythia-70M	L3	62.6% $\rightarrow$ 7.0%	61.9% $\rightarrow$ 7.9%	33.4%	355 $\rightarrow$ 415	0.338 $\rightarrow$ 0.184
	L6	8.2% $\rightarrow$ 4.6%	4.3% $\rightarrow$ 4.3%	2.0%	459 $\rightarrow$ 464	0.982 $\rightarrow$ 0.133

Takeaway. Sink strength moves the share, the share fixes the rank statistic through Equation 8, a synthetic sink reproduces the trajectory from a clean start, and a model without a sink shows no effect. The one feature of the reported trajectories the sink does not explain is the transient before step 1,000.

6 Scope across layers, scales, and architectures

6.1 Rogue dimensions and the leading direction are the same object

Timkey & van Schijndel (2021) identified a handful of coordinates that dominate cosine similarity in transformer representations and argued that they obscure representational quality. In GPT-2 the top five such coordinates carry 87.3 to 88.6 percent of the leading direction’s ℓ_1 mass at layers L3 through L10, against 3.8 percent at L0 and 20.9 percent at L1. The rogue dimensions of Timkey & van Schijndel (2021) and the attention sink of Xiao et al. (2024) are the same phenomenon in two coordinate systems. Table 5 gives the five summaries at the peak layer of two models.

6.2 Anisotropy responds to the sink but is not driven by it

Proposition 4 splits anisotropy into a sink term and a residual term without asserting which dominates. Measurement says the residual term does. At the peak-sink layers of the three Pythia models, removing the leading direction changes uncentered anisotropy from 0.252 to 0.196, from 0.277 to 0.260, and from 0.537 to 0.480. Those are changes of 22, 6, and 11 percent, against $32\times$ to $243\times$ for effective rank on the same matrices. Anisotropy also has dynamics the sink cannot supply: it peaks around step 16, when α is still between 1.0 and 1.2, and falls afterwards. We report anisotropy as sink-sensitive and not as sink-driven.

6.3 Raw CKA tracks one shared direction across ten layers

Proposition 5 predicts that two layers sharing one high-norm rank-one term have $\text{CKA} \rightarrow 1$. Figure 4 shows the prediction in GPT-2: raw CKA rates L2 through L11 as indistinguishable, and after the projection the same block shows a monotone decay in depth. The off-diagonal mean falls from 0.703 to 0.610 and the largest single change is -0.569 , at the L2-L11 pair that raw CKA rated most similar. In Pythia-70M the off-diagonal mean rises from 0.360 to 0.444, with the largest change $+0.474$ at L4-L6. The direction of the aggregate change is not stable across models; the layer-pair structure is what changes.

6.4 Larger models produce larger distortion figures, partly by construction

Table 6 reports the layer of maximum distortion for three larger models. Pythia-6.9B reaches $D = 733$, GPT-2-XL reaches 288, and Qwen2.5-1.5B reaches 532. Corollary 1 caps each at the residual rank in the last column, and the two 1.5B models differ in D by $1.8\times$ while differing in that cap by $2.1\times$. A headline that contamination worsens with scale is partly a statement about model width. What grows with scale independently of width is E_1 itself, which reaches 0.998 in Qwen2.5-1.5B at L2 with a spectral gap of 154 and a position-0 norm ratio of 113.

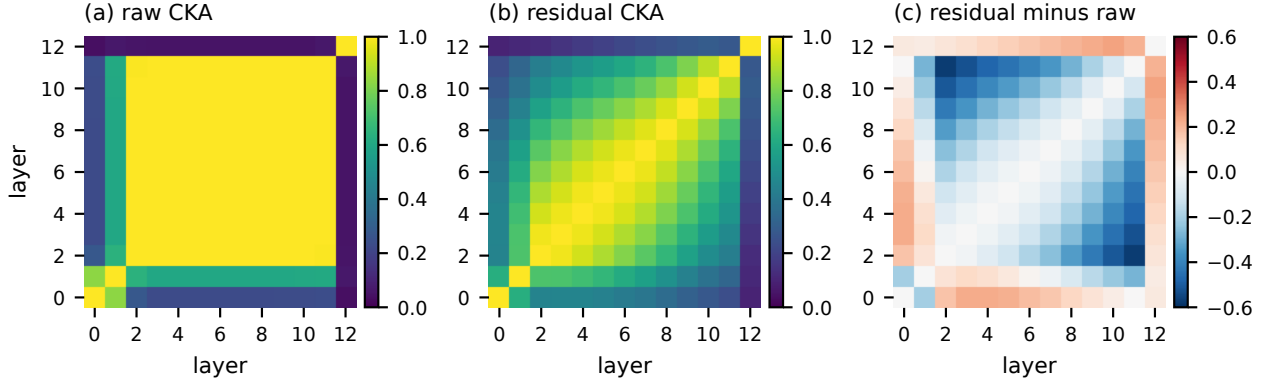


Figure 4: **Raw CKA reports one shared direction.** GPT-2, 22,819 WikiText-2 tokens. (a) Raw CKA marks L2 through L11 as indistinguishable at $\text{CKA} \geq 0.995$. (b) After the projection the same layers show a gradient in depth. (c) The difference. The largest single change is -0.569 between L2 and L11, the pair raw CKA rates most similar.

Table 6: **Contamination at the layer of maximum distortion, final checkpoints.** $D = \text{ER}(\text{C})/\text{ER}(\text{H})$, and the bound is Corollary 1. The two 1.5B models differ by $1.8\times$ in D and by $2.1\times$ in the bound.

Model	Arch.	Params	Layer	$E_1(\text{H})$	$E_1(\text{C})$	D	bound	$\text{ER}(\text{C})$
GPT-2-XL	GPT-2	1.5B	L18	0.942	0.088	288		518
Qwen2.5-1.5B	GQA	1.5B	L4	0.998	0.037	532		546
Pythia-6.9B	GPT-NeoX	6.9B	L8	0.949	0.024	733		1298

Takeaway. The effect appears in every architecture we tested, including grouped-query attention with rotary embeddings, and it is confined to the layers with a large spectral gap. Distortion factors should be reported next to the residual rank that bounds them.

7 What the leading direction does not do

The direction carries no probeable information where it dominates. Table 7 reports linear probing on Pythia-70M. At L3 and L4, where the leading direction holds 63 to 65 percent of the variance, removing it changes accuracy by 0.04 and 0.06 points. Across L0 through L5 the largest change is 0.24 points. A direction that holds most of the variance and none of the linearly decodable next-token information is a scaffold for the forward pass rather than a store of content, which agrees with Barbero et al. (2025) and Zhang et al. (2025).

L6, the output of the last block, is the exception and it is instructive. There the leading direction holds 0.4 percent of the variance, the spectral gap implied by Table 5 is 1.4, and removing it raises accuracy by 12.13 points. By the scope rule of Section 4 that direction is not the sink. We report the number because it shows the projection doing something real at a layer where the sink identification fails, and we do not attribute the gain to the sink.

The sink direction has no causal privilege at position 0. We measured perplexity on Pythia-410M after removing the leading direction from the position-0 residual stream at one layer at a time, and compared against removing a random direction of the same magnitude, over 5 trials and 12 evaluation sentences. Removing the leading direction raises perplexity by up to $49\times$ over a baseline of 20.4, averaging $17.8\times$ across the 24 layers. The matched-random control averages $42.9\times$ and exceeds the leading direction at 13 of 24 layers. Position 0 is fragile to perturbation of any direction, and this experiment does not establish that the sink direction in particular is load-bearing.

Table 7: **Removing the leading direction does not remove probeable information.** Pythia-70M, next-token identity prediction over the 500 most frequent types, 33,603 samples from 500 WikiText-2 passages, 3 seeds. “Random” is the mean over 5 random unit directions.

Layer	Sink var	Raw	Residual	Δ residual	Δ random
L0	1.5%	28.21 ± 0.03	28.40 ± 0.08	+0.19	−0.02
L1	2.5%	36.20 ± 0.25	35.96 ± 0.18	−0.24	+0.02
L2	7.0%	39.20 ± 0.30	39.06 ± 0.42	−0.14	−0.02
L3	64.7%	40.24 ± 0.28	40.29 ± 0.28	+0.04	−0.00
L4	62.7%	42.24 ± 0.18	42.31 ± 0.20	+0.06	+0.03
L5	25.3%	43.85 ± 0.06	43.74 ± 0.08	−0.11	−0.03
L6	0.4%	30.92 ± 0.49	43.05 ± 0.28	+ 12.13	+0.45

Table 8: **Evidence behind each claim.** Every number in the paper traces to one row. Data files and the script that regenerates the figures and `numbers.txt` ship with the submission.

Claim	Models	Corpus	Tokens	Ckpts
Identity, ordering (Fig. 1)	Pythia 160M/410M/1.4B, OLMo-2	OWT	$3 \times 200,704$	20, 13
Sink identification (Tab. 2)	same	OWT	$3 \times 200,704$	20, 13
Dynamics (Fig. 2, Fig. 3)	same	OWT	$3 \times 200,704$	20, 13
PC specificity (Tab. 3)	Pythia 160M/410M	OWT	200,704	final
Sample size (Tab. 9)	Pythia-1.4B	OWT	8K to 524K	3
Five metrics (Tab. 5)	GPT-2, Pythia-70M	OWT	163,840	final
CKA (Fig. 4)	GPT-2, Pythia-70M	WikiText-2	22,819	final
Scale (Tab. 6)	GPT-2-XL, Pythia-6.9B	WikiText-2	25,600	final
	Qwen2.5-1.5B	WikiText-2	200,000	final
Injection (Tab. 4)	Pythia-70M L3	WikiText-2	200,000	step 512
Probing (Tab. 7)	Pythia-70M	WikiText-2, 500 psg.	33,603	final
Position-0 perplexity	Pythia-410M	12 sentences	≈ 150	final

OWT: OpenWebText, SHA-256 prefix 66f376629b477962, 28,552 distinct token ids, no duplicated sequences.

Decisions that use relative layer ordering survive the contamination. On Pythia-6.9B we ranked layers by Block Influence, the cosine distance between a layer’s input and output residual stream, and pruned the 8 lowest. Pruning by raw scores gives perplexity 31.27 and pruning by residual scores gives 33.81, against 16.23 for the full model. The two sets overlap in 7 of 8 layers. Distilling GPT-2 into a 6-layer student gives perplexity 5.57 for even layer spacing, 6.31 for raw CKA matching, and 6.01 for residual CKA matching. Cross-model CKA between GPT-2 and Pythia-70M rises from a maximum of 0.016 to 0.116 after the projection, so the small cross-model similarity is not an artifact of a shared sink. The contamination distorts absolute values of spectral summaries and leaves the ordering of layers largely intact.

Takeaway. The leading direction dominates the spectrum, carries no probeable next-token information, and is not more causally privileged at position 0 than a random direction of the same size. Its effect is on measurement.

8 Scope and limitations

Effective rank has not converged in sample size. Table 9 resamples Pythia-1.4B L5 at four corpus sizes. E_1 , α , and anisotropy settle to within 2 percent by $n = 524,288$; effective rank and RankMe do not, and both rise monotonically with n . At the final checkpoint $\text{ER}(\mathbf{C})$ moves from 756 to 1036 over the four sizes. This is the sample-spectrum bias of a 2048-dimensional covariance estimated from finitely many tokens: small eigenvalues are underestimated at low n , so an entropy-based rank approaches its limit from below. Two consequences. Absolute effective ranks in this paper are values at a stated n and not properties of a layer. Comparisons are valid because every point in a given figure uses the same n , and the raw-to-residual ratio at step 143,000 is $194\times$ at $n = 8,192$ and $246\times$ at $n = 524,288$, so the conclusion does not turn on the choice.

Table 9: **Half of the summaries still move with sample size.** Pythia-1.4B L5, one block resampled at four sizes. Entries are the largest deviation from the value at $n = 524,288$. Ratio-type quantities converge; entropy-based ranks do not, and rise monotonically with n .

	step 512	step 8,000	step 143,000
E_1 (raw)	8.9%	4.8%	0.5%
α	2.0%	4.2%	0.8%
anisotropy (raw)	16.0%	2.7%	1.6%
ER(H)	13.0%	23.4%	7.7%
ER(C)	12.8%	23.5%	27.1%
RankMe(H)	11.1%	8.9%	11.5%

The OLMo-2 control covers 2 percent of a run. Step 36,000 is 76B tokens. Pythia’s sink at these layers becomes visible only after a comparable fraction of its schedule, and OLMo-2’s α is rising at the last measured checkpoint. Section 5 claims that OLMo-2 has no sink over the measured interval, which is what a control needs, and not that OLMo-2 never forms one.

One protocol change across the paper. The dynamics results use 200,704 OpenWebText tokens per block. The GPT-2, Pythia-70M, Pythia-6.9B, and Qwen2.5 measurements come from an earlier round with 22,819 to 200,000 tokens on WikiText-2 or OpenWebText, listed per row in Table 8. Given Table 9, effective ranks are not comparable across those rows, and we compare only within a row.

The position-0 perplexity experiment is small. It uses 12 hand-written sentences on one model, which is enough to show that the matched-random control is not smaller than the treatment and not enough to characterize the causal role of the sink.

The projection is linear and applies at one layer at a time. We remove one direction from one layer and recompute summaries at that layer. We do not track how the removal propagates through subsequent layers, and we do not claim that a nonlinear correction would give the same residual.

The identities assume the leading direction is removed. Equations 8 and 9 hold when the projected direction is the top eigenvector. Table 4 shows what happens when it is not: below $\alpha = 3$ the prediction fails by 8 percent. In practice we check $\sigma_1/\sigma_2 > 3$ before reading **s** as the sink.

We do not evaluate whether phases exist. Our result is that the metrics used to detect them are determined by one direction. A study that reports phases in the residual quantities is unaffected by anything here.

9 Related work

Spectral summaries as descriptions of training. Li et al. (2025) track RankMe and a spectral exponent across pretraining and post-training in OLMo and Pythia and report three phases. Garrido et al. (2023) introduced RankMe as a label-free proxy for downstream quality in self-supervised learning, Roy & Vetterli (2007) defined effective rank, Ethayarajh (2019) measured anisotropy in contextual embeddings, and Raghu et al. (2021) used CKA to compare vision transformers with convolutional networks. Kulkarni et al. (2026) trained 108 models and found that effective rank tracks batch size and weight decay more than it tracks performance, a confound in the training configuration rather than in the estimator. Our result is about the estimator. It also says which of these statistics is the safer choice. RankMe absorbs a 1.2 to 1.5 factor from the sink where effective rank absorbs 32 to 243, so a study that must report one number should report RankMe together with its residual.

Attention sinks. Xiao et al. (2024) named the attention sink and used it for streaming inference; Darcet et al. (2024) added register tokens to vision transformers for the same reason. Sun et al. (2024) characterized

massive activations and Gu et al. (2025) showed that sinks emerge early and strengthen with scale. Barbero et al. (2025) argued that sinks limit over-mixing, Cancedda (2024) related sink signals to the unembedding spectrum, and Zhang et al. (2025) showed that one or two sink tags explain a large share of PCA variance in attention outputs. Table 2 adds the trajectory of the position-0 mass of the leading eigenvector, which starts at 0.1 percent and reaches half.

The closest prior result. Queipo-de Llano et al. (2025) prove that massive activations force effective rank down at a layer, and read the resulting compression valley as a functional stage of the forward pass. Proposition 2 sharpens that from an inequality at one layer to an identity, and Lemma 2 adds which statistics inherit it. The consequence for training-dynamics measurement follows: given the share and the residual rank, effective rank has no remaining degree of freedom to carry. Where Queipo-de Llano et al. (2025) ask what sinks do inside a forward pass, we ask what they do to a measurement taken across checkpoints.

Removing a dominant direction. Mu & Viswanath (2018) remove the top principal components of word embeddings and improve similarity benchmarks; Li et al. (2020) and Rajaei & Pilehvar (2021) pursue the same goal with normalizing flows and clustering. Those papers optimize a downstream score. We use the same operation as a control and report the residual next to the raw value rather than in place of it. Timkey & van Schijndel (2021) identified the rogue dimensions that Section 6 matches to the leading direction.

10 Conclusion

Effective rank and RankMe factor exactly into a binary-entropy term and a power of the residual value, so a study that reports either has reported a share and a residual. The share differs between the two statistics, never in RankMe’s disfavour, and that ordering decides the damage. At the peak-sink layer of Pythia-1.4B one direction holds 85 percent of the eigenvalue mass and 6 percent of the singular-value mass; the resulting distortions are $243\times$ and $1.24\times$. The direction is the attention sink. It reaches half of its energy on the 0.2 percent of tokens at position 0, and its strength alone accounts for 94 to 97 percent of the variance in E_1 . A model that has not yet formed one shows raw and residual metrics agreeing to 3.5 percent. After the projection, the $105\times$ collapse in effective rank is gone, the residual grows by a factor of two, and one transient before step 1,000 remains.

Reporting practice follows. A paper that quotes effective rank should quote E_1 and $ER(C)$ beside it, since those two determine it, and should quote the spectral gap so a reader can tell whether the removed direction is the sink. A paper that quotes a distortion factor should quote the residual value that bounds it, and the token count, since entropy-based ranks are not converged at half a million tokens. A paper that wants one rank statistic should prefer RankMe and still report its residual, because a $1.24\times$ offset was enough to invert the sign of two of our three trajectories. The one-line implementation is `H_res = H - (H @ s)[: , None] * s`, with `s` the top right singular vector of the centered `H`.

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A Proofs

A.1 Lemma 1

Proof. Write $\tilde{w}_i = w_i/\|\mathbf{w}\|_1$, so $\tilde{w}_1 = t$ and $\sum_{i \geq 2} \tilde{w}_i = 1-t$. Let $q_i = \tilde{w}_i/(1-t)$ for $i \geq 2$ be the normalization of $\mathbf{w}_{-1}$. Then

$$-\sum_i \tilde{w}_i \log \tilde{w}_i = -t \log t - \sum_{i \geq 2} (1-t) q_i [\log(1-t) + \log q_i] = h_b(t) + (1-t) \left(-\sum_{i \geq 2} q_i \log q_i \right),$$

using $\sum_{i \geq 2} q_i = 1$. Exponentiating gives Equation 6. The case $t = 1$ makes both sides equal 1 under the convention $0 \log 0 = 0$. $\square$

A.2 Lemma 2

Proof. Write $S_1 = \sum_i \sigma_i$ and $S_2 = \sum_i \sigma_i^2$. Then $p = \sigma_1^2/S_2$ and $r = \sigma_1/S_1$, so $p/r = \sigma_1 S_1/S_2$. Since $\sigma_i \leq \sigma_1$ for every i ,

$$S_2 = \sum_i \sigma_i^2 \leq \sigma_1 \sum_i \sigma_i = \sigma_1 S_1,$$

which gives $p/r \geq 1$. Equality requires $\sigma_i(\sigma_1 - \sigma_i) = 0$ for every i , that is, every nonzero singular value equals σ_1 ; combined with σ_1 being the largest, the spectrum is a single repeated value, and $p = r$ then forces one nonzero entry. $\square$

A.3 Spectrum of the residual

Lemma 3. *With $\mathbf{C}$ as in Equation 5 and $\mathbf{s} = \mathbf{v}_1$, the covariance of $\mathbf{C}$ is $\mathbf{\Sigma} - \lambda_1 \mathbf{s} \mathbf{s}^\top$ and its eigenvalues are $\lambda_2 \geq \dots \geq \lambda_d \geq 0$ together with one extra zero.*

Proof. Centering is left multiplication by $\mathbf{I}_n - \frac{1}{n} \mathbf{1} \mathbf{1}^\top$ and the projection is right multiplication by $\mathbf{I}_d - \mathbf{s} \mathbf{s}^\top$, so the two commute and $\bar{\mathbf{C}} = \bar{\mathbf{H}}(\mathbf{I}_d - \mathbf{s} \mathbf{s}^\top)$. Hence

$$\frac{1}{n} \bar{\mathbf{C}}^\top \bar{\mathbf{C}} = (\mathbf{I} - \mathbf{s} \mathbf{s}^\top) \mathbf{\Sigma} (\mathbf{I} - \mathbf{s} \mathbf{s}^\top) = \mathbf{\Sigma} - \lambda_1 \mathbf{s} \mathbf{s}^\top,$$

where the second equality uses $\mathbf{\Sigma} \mathbf{s} = \lambda_1 \mathbf{s}$ and $\mathbf{s}^\top \mathbf{s} = 1$. In the eigenbasis of $\mathbf{\Sigma}$ this matrix is diagonal with entries $0, \lambda_2, \dots, \lambda_d$. $\square$

A.4 Propositions 1, 2, 3 and Corollary 1

Proof. By Lemma 3, $\text{tr}(\mathbf{\Sigma} - \lambda_1 \mathbf{s} \mathbf{s}^\top) = V - \lambda_1$, so $x = \lambda_1/(V - \lambda_1)$ and $x/(x+1) = \lambda_1/V = E_1(\mathbf{H})$, which proves Proposition 1; the map $x \mapsto x/(x+1)$ is strictly increasing with inverse $p \mapsto p/(1-p)$.

For Proposition 2 apply Lemma 1 to $\mathbf{w} = (\lambda_1, \dots, \lambda_d)$. Its largest entry is λ_1 and $t = \lambda_1/V = E_1(\mathbf{H}) = p$. The deleted vector $\mathbf{w}_{-1} = (\lambda_2, \dots, \lambda_d)$ is the nonzero part of the spectrum of the residual covariance, and appending the extra zero of Lemma 3 leaves H_{exp} unchanged, so $H_{\text{exp}}(\mathbf{w}_{-1}) = E(\mathbf{C})$. Substituting into Equation 6 gives Equation 8. Proposition 3 is the same argument with $\mathbf{w} = (\sigma_1, \dots, \sigma_d)$ and $t = \sigma_1/\sum_i \sigma_i$, since $\sigma_i = \sqrt{n \lambda_i}$ preserves the ordering and the residual singular values are $\sigma_2, \dots, \sigma_d$.

For Corollary 1, divide: $D_M = M(\mathbf{C})/M(\mathbf{H}) = M(\mathbf{C})^{1-(1-t)} e^{-h_b(t)} = M(\mathbf{C})^t e^{-h_b(t)}$. Both factors are at most $M(\mathbf{C})$ and 1 respectively for $t \in [0, 1]$ and $M(\mathbf{C}) \geq 1$, so $D_M \leq M(\mathbf{C})$, with equality in the limit $t \rightarrow 1$ where $h_b(t) \rightarrow 0$. Taking logarithms, $D_M > 1$ if and only if $t \log M(\mathbf{C}) > h_b(t)$, and $\partial_t \log D_M = \log M(\mathbf{C}) - \log \frac{1-t}{t}$ is positive whenever $D_M > 1$, so D_M increases in t on that range. $\square$

A.5 Proposition 4

Proof. Let $\hat{\mathbf{u}}_i = \mathbf{H}_i/\|\mathbf{H}_i\|$ and $S = \sum_i \hat{\mathbf{u}}_i$. Then $\|S\|^2 = \sum_{i,j} \langle \hat{\mathbf{u}}_i, \hat{\mathbf{u}}_j \rangle = n + \sum_{i \neq j} \langle \hat{\mathbf{u}}_i, \hat{\mathbf{u}}_j \rangle$, and $\rho(\mathbf{H})$ is the average of the off-diagonal terms, which gives the first equality. Decomposing $\hat{\mathbf{u}}_i = \beta_i \mathbf{s} + \hat{\mathbf{u}}_i^\perp$ with $\langle \mathbf{s}, \hat{\mathbf{u}}_i^\perp \rangle = 0$

gives $\|S\|^2 = (\sum_i \beta_i)^2 + \|\sum_i \hat{\mathbf{u}}_i^\perp\|^2$, the second equality. If $\beta_i = \beta$ for all i then the first term is $n^2\beta^2$ and $\rho \geq (n^2\beta^2 - n)/(n(n-1)) \rightarrow \beta^2$ for large n . $\square$

A.6 Proposition 5

Proof. Since $\mathbf{X}_c^\top \mathbf{u} = \mathbf{0}$ and $\mathbf{Y}_c^\top \mathbf{u} = \mathbf{0}$, the cross terms vanish and $\mathbf{X}^\top \mathbf{Y} = \|\mathbf{u}\|^2 \mathbf{p}\mathbf{q}^\top + \mathbf{X}_c^\top \mathbf{Y}_c$. Write $c_{xy} = \|\mathbf{X}_c^\top \mathbf{Y}_c\|_F$, $c_{xx} = \|\mathbf{X}_c^\top \mathbf{X}_c\|_F$, $c_{yy} = \|\mathbf{Y}_c^\top \mathbf{Y}_c\|_F$, and note $\|\mathbf{p}\mathbf{q}^\top\|_F = \|\mathbf{p}\| \|\mathbf{q}\|$, so $\|\|\mathbf{u}\|^2 \mathbf{p}\mathbf{q}^\top\|_F = \sqrt{AB}$. The triangle inequality gives

$$\|\mathbf{X}^\top \mathbf{Y}\|_F^2 \geq (\sqrt{AB} - c_{xy})^2, \quad \|\mathbf{X}^\top \mathbf{X}\|_F \leq A + c_{xx}, \quad \|\mathbf{Y}^\top \mathbf{Y}\|_F \leq B + c_{yy}.$$

Hence

$$\text{CKA}(\mathbf{X}, \mathbf{Y}) \geq \frac{AB(1 - c_{xy}/\sqrt{AB})^2}{AB(1 + c_{xx}/A)(1 + c_{yy}/B)} = 1 - O(A^{-1/2} + B^{-1/2}),$$

expanding to first order in $1/\sqrt{A}$ and $1/\sqrt{B}$. Since $\text{CKA} \leq 1$ always, the two-sided statement follows. $\square$

A.7 Proposition 6

Proof. Replacing $\mathbf{s}$ by $\mathbf{v}_k$ in Lemma 3 gives a residual covariance with spectrum $\{\lambda_i\}_{i \neq k}$ plus a zero, whose largest entry is λ_1 for $k \geq 2$ and whose trace is $V - \lambda_k$. Hence $E_1(\mathbf{H}_{-k}) = \lambda_1/(V - \lambda_k) \geq \lambda_1/V$. For $\mathbf{u}$ uniform on the sphere, $\mathbb{E}[\mathbf{u}\mathbf{u}^\top] = \mathbf{I}_d/d$, so $\mathbb{E}[\mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u}] = \text{tr}(\boldsymbol{\Sigma})/d = V/d$. $\square$

B Additional measurements

Cross-block spread. Median relative standard deviation over the three disjoint 200,704-token blocks, at the peak-sink layers, across all 20 checkpoints: $\text{ER}(\mathbf{H})$ 0.98 to 1.18 percent, $\text{ER}(\mathbf{C})$ 0.73 to 0.87 percent, E_1 1.06 to 1.86 percent, $\text{RankMe}(\mathbf{C})$ 0.08 to 0.16 percent, α 0.20 to 0.25 percent.

Spectral gaps by layer. GPT-2: 1.1, 3.7, 14.7, 14.9, 14.6, 13.4, 12.4, 9.6, 8.4, 5.5, 3.2, 3.0 at L0 through L11. Qwen2.5-1.5B reaches 154.2 at L2 with a position-0 norm ratio of 112.8.

Per-layer contamination at scale. Pythia-6.9B, raw and residual E_1 by layer: L0 0.165/0.035, L4 0.931/0.030, L8 0.949/0.024, L12 0.929/0.023, L16 0.901/0.025, L20 0.853/0.036, L24 0.777/0.047, L28 0.700/0.051, L31 0.075/0.059. The first and last layers are uncontaminated in every model we measured.

Pythia-70M under the earlier protocol. WikiText-2, 22,851 tokens, one block. L3 effective rank falls from 178 to 9.2 raw and moves from 191 to 156 in the residual, with a maximum of 236 at step 4,000. L1, which never develops a sink, moves from 187 to 162 raw and 202 to 190 in the residual. Table 9 shows that effective rank at $n = 22,851$ is far from its large- n value, so we do not compare these numbers with the $n = 200,704$ trajectories of Section 5.

Full injection sweep. Table 4 omits $\alpha = 50$, where $E_1 = 0.961$ and $\text{ER} = 1.43$ against 1.43 predicted by Equation 8.

C Reproduction

The submission ships the experiment scripts, the raw output of every run quoted in the paper, and `build.py`, which regenerates all four figures and writes `numbers.txt` containing every quantity in the manuscript next to the file it came from. Running `python build.py` takes under a minute and requires no GPU. The measurements use public checkpoints and total under 40 GPU-hours.

An earlier round of our own experiments measured Pythia-160M and Pythia-1.4B training dynamics through a corpus loader whose exception handler silently substituted 200 copies of one template sentence. The

resulting effective ranks were near 10 at initialization for a 768-dimensional model, which is the signature of roughly ten distinct token types. Those runs are excluded, every number derived from them is withdrawn, and Sections 4 and 5 report the replacement measurements on the corpus described in Section 2. The corpus is published with its SHA-256 prefix, its distinct-token count, and a check for duplicated sequences; the shipped scripts raise on a corpus-loading failure and refuse to run when a measured effective rank at initialization falls below $d/25$.

Broader Impact Statement

This work identifies a measurement bias in a class of published analyses of transformer training. Its effect is on the accuracy of scientific claims about representation geometry. We see no direct societal risk from the method or from its release.